Malav Patel

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Skills

Languages: Python, C, C++, CUDA, Julia, MATLAB
Frameworks: PyTorch, Tensorflow, JAX, OpenCV, Docker, Kubernetes
Research Topics: GAN, VAE, DDPM, Normalizing Flows, Vector Quantization, Neural Audio Codecs, RNNs, GPT, BERT, RLHF, Mixture of Experts, Classifier (Free) Guidance, Score Matching, Speculative Decoding, Variational Inference, Stochastic Differential Equations, Rectified Flows

Experience

Graduate Research Assistant, Georgia Institute of Technology – Atlanta, GA — August 2022 – Present
- Applying gradient-based methods, integer programming, and linear programming to optimize placement and tasking of satellite constellations in cislunar space.

AI Trainer, Data Annotation – Atlanta, GA — June 2023 – Present
- Engineering difficult prompts and problems to challenge SOTA multi-modal models across english, mathematics, and science tasks.
- Evaluating reasoning and agentic models to improve instruction following, correctness, and verbosity.

Intern, Chamberlain Group – Chicago, IL — June 2021 – September 2021
- Trained a sequence model on time-series data on gearboxes to identify manufacturing faults in rotary mechanims, achieving 88% accuracy.

Projects

On Device GPT Inference — github.com/Malav-P/staticgrad
- Built a library in C++ for running training and inference of GPT2.
- Optimized inference time on Apple M1 Hardware using Apple's Accelerate GEMM kernels and custom attention kernel, achieving inference time latency of 25 ms/token without compiler optimizations.

Open SoundStream — malav-p.github.io/soundstream
- Established an open source implementation of SoundStream, a neural audio codec underpinning SOTA generative audio models in 2 weeks.
- Trained on a single NVIDIA L40S GPU in less than 15 hours. Released checkpoints for other users.

ModernBERT for Patent Classification — github.com/Malav-P/modernpatentBERT
- Trained modernBERT on the patent classification task with >2x speedup in training throughput achieving SOTA precision and F1 scores on a domain test set.
- Hosted an open repository on Huggingface of 3 million patents for future work on the classification task.

CNNs in C++ — github.com/Malav-P/CNN
- Built a C++ header-only library for constructing, training, and running convolutional neural networks.
- Achieved hand-written digit recognition at 60 Hz on Apple M1 with a CNN trained on MNIST.
- Implemented custom matmul kernels in CUDA to parallelize intensive matrix multiplication operations, achieving a 3x speedup on matmul calls.

Education

Georgia Institute of Technology – PhD in Aerospace Engineering — June 2026
Georgia Institute of Technology – MS in Computer Science — December 2025
Northwestern University – BS in Physics, Mechanical Engineering, Integrated Sciences — June 2022